HireSense

Software Requirements Specification

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Revision History

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1.0	27 Oct 2026	Sohail Ahmed		Initial full-project SRS prepared for review.

Document control

Baseline: HireSense project repository, 27 October 2026. Formal review and acceptance are recorded by the project team and supervisor; this version does not assert completed acceptance testing.

Reading this specification

Sections 1 and 2 establish scope and context. Section 3 defines the functional requirements. Sections 4 and 8 define screens and interaction behavior. Sections 5 to 7 address quality and design constraints. Section 9 covers data, operations, traceability and end-to-end acceptance.

Requirement convention

Shall identifies a mandatory acceptance requirement. FR, NFR, PER, SAF, SEC, QUA, OTH and UI/UX identifiers are unique within this document. A linked TC identifier defines a planned verification scenario, not a statement that the scenario has passed.

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1 Introduction

1.1 Purpose

This Software Requirements Specification defines the functional, interface, quality, safety and operational requirements of HireSense, an AI-assisted recruitment platform. It provides a common baseline for the project team, supervisor, reviewers and testers to understand the system and evaluate its behavior.

This specification establishes the acceptance baseline for the current HireSense project. Requirements do not certify that every scenario has already passed. Performance targets in Section 5.1 require controlled measurement. Team details and formal approval are to be completed separately.

1.2 Scope

HireSense covers the recruitment flow from preparing and approving a job description through publication, application intake, CV screening, interview scheduling, live interviews, transcript assessment, a short practical exercise and evidence-backed final ranking. It supports technical and nontechnical roles.

The primary user is an authenticated interviewer. Candidates apply through Google Forms and participate through private invitation links without a HireSense account. AI assists drafting, retrieval, analysis and task review; application services enforce authorization, state transitions and score arithmetic. The interviewer remains responsible for employment decisions.

In scope	Outside this baseline
RAG/manual JDs, version approval and confirmed Discord publication.	General-purpose job boards, an applicant marketplace or autonomous job publication.
Read-only Sheets/Drive intake; PDF/DOCX CV screening and scanned-PDF OCR.	Writing to source forms, automatic OCR of all knowledge PDFs or verifying every CV claim externally.
Reviewed SMTP scheduling, private links, ICS and LiveKit interviews.	Google/Outlook calendar booking, group interviews or interview A/V recording.
Consented transcripts, job-related analysis and traceable candidate summaries.	Candidate-visible transcripts, personality/emotion inference or guaranteed perfect ASR/AI grading.
One short live work sample; optional local photo comparison.	Take-home attempts, executing submitted code, plagiarism/AI-use detection or legal-ID/liveness assurance.
Saved-score final ranking and private reports.	Automatic hiring/rejection, payroll, onboarding or a separate interview-quality-rating module.

1.3 Intended Stakeholder

Stakeholder	Responsibilities / use
Interviewer / recruiter	Prepare jobs; review AI drafts and evidence; select candidates; confirm publication/invitations; conduct interviews; approve practical reviews; make decisions.
Candidate	Submit an application; join the intended interview; choose optional permissions; answer questions; submit one live practical response.
Project team / maintainer	Implement, configure, migrate and maintain the platform; investigate safe diagnostics and provider failures.
Supervisor / evaluator	Review scope, innovation, architecture and acceptance evidence against the agreed project requirements.
Deployment operator / data owner	Provision accounts, credentials and models; manage permissions, retention, backups and external-provider obligations.

1.4 References

Repository references are relative to the project root. They support this specification but do not replace its acceptance criteria. External service details are configuration dependencies, not service-level guarantees.

Ref. ID	Document / source	Use
REF01	SRS Template.doc (user-supplied)	Section order, document control and requirement-table structure.
REF02	Final_Project_doc_Business_Info_Document_AI_Recruitment_Platform_v2.pdf	Business context, core recruitment capabilities and stretch/innovation context.
REF03	README.md; docker-compose.yml; backend/app/config.py	Project scope, deployed components, configuration and limits.
REF04	docs/diagrams/High_Level_Diagram.drawio; docs/diagrams/HireSense_System_Architecture.drawio	High-level and detailed architecture views.
REF05	docs/langgraphs/README.md; docs/system-prompts.md	Active graph inventory, tool scopes and prompt responsibilities.

Ref. ID	Document / source	Use
REF06	docs/robust-cv-screening.md; docs/cv-ocr.md; docs/deterministic-candidate-scoring.md	CV processing, evidence validation and deterministic arithmetic.
REF07	docs/interview-scheduling.md; docs/interview-rooms.md; docs/live-transcription.md	Scheduling, room lifecycle and consented speech processing.
REF08	docs/interview-analysis.md; docs/live-work-samples.md; docs/consolidated-assessments.md	Answer marks, practical review and saved-score final ranking.
REF09	docs/candidate-identification.md; docs/account-access.md; docs/workflow-navigation.md	Optional identity comparison, access boundaries and UI transitions.
REF10	docs/llm-token-usage-estimate.md; backend/tests/; frontend/tests/	Workload estimates and implementation verification areas.

1.5 Definitions Acronyms and Abbreviations

Term	Meaning in HireSense
Application	One candidate submission associated with a specific job posting and source response identity.
ASR / STT	Automatic speech recognition / speech-to-text; may contain errors or missing speech.
CV	Curriculum vitae / résumé submitted as a supported document.
Evidence	Exact source text plus an application-specific locator, version and provenance used to support a finding.
JD	Job description and its approved screening requirements.
RAG	Retrieval-augmented generation: relevant authorized passages supplied to generation.
LLM	Large language model used for generation and evidence-supported assessment.
Embedding / pgvector	A numeric text representation / PostgreSQL extension for vector storage and retrieval.
MCP	Model Context Protocol; private services expose job posting and scheduling operations.

Term	Meaning in HireSense
LangGraph	Stateful workflow orchestration; seven active graph definitions are listed in Section 6.
Approved guide	An authorized answer guide/reference used for correctness, not a candidate CV or another candidate's answer.
Compatible result	A saved validated assessment matching the current relevant source hashes, configuration and policy versions.
ICS / SMTP	Calendar invitation file format / protocol used to deliver invitation email.
WebRTC / LiveKit	Browser real-time audio/video transport / the configured room-media provider.
FR / TC / p95	Functional requirement / acceptance test scenario / 95th percentile of a measured sample.

2 Overall Description

2.1 Overview

HireSense is a web application with a modular backend, persistent recruitment records and scoped AI workflows. The browser handles user interaction and media. Backend services own recruitment state, evidence validation, access control and exact score calculation. Separate integration and transcription processes support selected external operations.

2.1.1 End-to-end recruitment flow

Stage	Main interaction	Persisted result / gate
1 Prepare	Upload reference knowledge; describe the role or create a manual JD.	Draft, sources and screening profile; explicit version approval.
2 Publish	Review the Discord destination, application link, deadline and content.	Confirmed publication and external reference.
3 Import	Sync Google Form response rows and private Drive documents.	Application-scoped profile, source identity, CV and optional photo.
4 Screen	Extract/OCR CV; retrieve and validate requirement evidence.	Validated findings, configuration snapshot and CV ranking; failures remain visible.
5 Schedule	Select eligible candidates; preview slots, email and ICS; confirm.	Interview, invitation outcome, private access and schedule history.
6 Interview	Device checks, admission, private call and optional permissions.	Room state; final consented transcript text, not A/V recordings.
7 Assess	Analyze saved answers; run one short task and approve its verified AI review.	Interview marks, performance summary and approved practical score.
8 Rank	Calculate using current compatible saved components; inspect/export.	Final ranking and private evidence-linked report; human decision.

2.1.2 System boundary and external dependencies

Component / dependency	Responsibility	Boundary / failure behavior
Web frontend and FastAPI	Interviewer/candidate APIs, state, validation and UI.	Private resources require role/ownership scope.
PostgreSQL + pgvector	Recruitment data, provenance, workflow records and vectors.	Not a shared public vector database.

Component / dependency	Responsibility	Boundary / failure behavior
Ollama + nomic-embed-text	Local/container embedding generation.	Initialization and model identity must pass before dependent work.
Configured Gemini service	JD generation, tool routing, evidence analysis and AI reviews.	Internet, valid credentials and quota required; failures are not scores.
Google Sheets / Drive	Read-only application intake and authorized file retrieval.	No form/source deletion or write capability is assumed.
Discord + job-posting MCP	Confirmed job publication.	External effects occur only through the reviewed operation.
SMTP + scheduling MCP	Invitation/reschedule delivery and calendar attachments.	Uncertain sends require reconciliation, not blind retry.
LiveKit Cloud	Private one-to-one WebRTC media and scoped room tokens.	Media connects directly, independently of a website tunnel.
Deepgram + transcription worker	Optional live microphone transcription.	Both consents and interviewer start; provider loss does not end the call.
Local OCR and identity models	Scanned CV text; optional one-to-one face comparison.	Models/limits are locally configured; no face data is sent to the LLM.

2.1.3 User characteristics and assumptions

Interviewers use a modern desktop browser and can review job criteria and evidence. Candidates need a supported browser, a private link and suitable microphone/camera permissions. A narrow screen may reflow the interface; desktop side-by-side layout is the primary interview-console experience.

An operator provisions the interviewer account, persistent storage, database, integration permissions, service credentials and model assets. Internet connectivity is required for cloud services and initial model downloads. Containerized embedding removes the need for a separately configured host Ollama installation, but does not make the complete system offline or configuration-free.

Application documents and reference guides are assumed to be lawfully supplied by their owners. Optional permissions remain voluntary. Model findings may be wrong: uncertainty, source gaps and verification needs must be shown rather than concealed. Photo matching is an optional aid only.

2.1.4 Score and state model

All component scores use a 0–100 percentage scale. CV evidence labels inform deterministic requirement coverage; interview and practical numeric marks are evidence-supported AI assessments. Their labels are not interchangeable scoring formulas.

Score type	Policy	Incomplete behavior
CV	Met = 1, partial = 0.5, unmet = 0; approved requirement/category weights. Defaults: skills 50%, experience 30%, qualifications 20%.	Unreadable, failed or stale evidence is unranked.
Interview	Four equally weighted dimensions: relevance, accuracy, completeness and quality. Average validated included-question scores.	Housekeeping excluded; verified mismatch/unanswered = 0; uncertain/failed analysis remains incomplete.
Practical	Verified AI correctness result, then explicit interviewer approval of that numerical result.	Unsubmitted, unverified, failed or unapproved results do not contribute as zero.
Final	Interview 60% + CV 30% + approved practical 10%. Exact totals govern ties.	All required components must be current and compatible; weights are not silently redistributed.

Example: interview 80, CV 70 and practical 90 produce a final score of 78.00%. Display percentages round half-up to two decimals; exact totals determine shared ranks, for example 1, 1, 3. Equivalent-quality candidates may legitimately tie.

2.2 Technical platform

Layer	Technology / execution	Requirement implication
Client	React 19, TypeScript, Vite, Tailwind CSS; Nginx delivery.	Browser UI calls the backend through /api; no provider keys in the client.
Application	Python 3.12, FastAPI, Pydantic, SQLAlchemy and Alembic.	Validated API contracts, shared services and versioned DB migrations.
Persistence	PostgreSQL 16 with pgvector; persistent private storage.	Owned relational state, 768-dimension vectors and protected file/model volumes.
AI orchestration	LangGraph / LangChain; configured Gemini generation; Ollama embeddings.	Scoped tools and bounded recovery, not autonomous recruitment authority.

Layer	Technology / execution	Requirement implication
Documents and media	Native PDF/DOCX extraction, Poppler/Tesseract, isolated report browser, YuNet/SFace.	Local processing is bounded and source-aware; submitted code is never executed.
Runtime	Docker Compose; backend, frontend, MCP services, transcription worker, database and Ollama.	One-shot ollama-init prepares/validates the model; persistent cache avoids repeated downloads.
Demo tools	Mailpit and Cloudflare demo profile.	Mailpit is a mail sink; a public tunnel needs explicit safe deployment and synthetic demo data.

Background work uses in-process asynchronous/threaded execution with persisted run states and recovery handling. This baseline does not include Redis, Celery, a distributed job cluster or a candidate-account subsystem.

3 Functional Requirements

3.1 Overview

Each feature below identifies its actors, purpose and mandatory behavior. The requirement table supplies an observable acceptance scenario and limitations. FRxx.yy identifies a requirement; TCxx.yy identifies its corresponding planned acceptance scenario. Test IDs are specification identifiers, not reports of executed tests.

No.	Capability	Requirements
01	Interviewer authentication and candidate access	FR01.01 to FR01.05
02	Knowledge base and embedding retrieval	FR02.01 to FR02.04
03	Job description preparation and approval	FR03.01 to FR03.05
04	Job publication through Discord	FR04.01 to FR04.04
05	Application intake and candidate management	FR05.01 to FR05.06
06	CV extraction OCR and evidence screening	FR06.01 to FR06.06
07	CV scoring and ranking	FR07.01 to FR07.05
08	Interview scheduling invitations and rescheduling	FR08.01 to FR08.06
09	Live interview rooms and participant consoles	FR09.01 to FR09.05
10	Consented live transcription	FR10.01 to FR10.05
11	Central and private interview assistants	FR11.01 to FR11.05
12	Transcript analysis and candidate performance	FR12.01 to FR12.07
13	Live Work Sample Studio	FR13.01 to FR13.07
14	Optional photo verification and candidate profile	FR14.01 to FR14.05
15	Final score consolidation and ranking	FR15.01 to FR15.05
16	Candidate reports evidence and exports	FR16.01 to FR16.04
17	Dashboard notifications and workflow navigation	FR17.01 to FR17.04

3.1.1 Interviewer authentication and candidate access

Actors: Interviewer, candidate. The system separates authenticated interviewer workspaces from narrowly scoped candidate participation.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR01.01	The system shall authenticate active interviewers using hashed passwords and signed access tokens.	Valid credentials grant interviewer access; invalid credentials return a generic denial.	No public signup or candidate account.	TC01.01
FR01.02	The system shall authorize every protected resource against its owning interviewer.	A second interviewer cannot read, change, export or delete another owner's records.	Backend checks apply to API and chat.	TC01.02
FR01.03	The system shall exchange expiring private interview links for scoped HttpOnly candidate sessions.	A valid link opens only its intended interview; expired, revoked or altered links are rejected.	HTTPS secure-cookie settings are deployment dependent.	TC01.03
FR01.04	The system shall prevent candidate access to private interviewer information.	Candidate requests for transcripts, analysis, assistant messages, match outcomes and reports are denied.	Hiding controls alone is insufficient.	TC01.04
FR01.05	The system shall revoke obsolete access when invitation links change or interviews close.	Old links cannot issue new access after revocation or closure.	Previously issued sessions must respect room state.	TC01.05

3.1.2 Knowledge base and embedding retrieval

Actors: Interviewer, background services. Typed reference knowledge supports drafting and interview questions. Private candidate passages are not shared reference material.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR02.01	The system shall ingest authorized reference PDFs with document, page, chunk and source-type metadata.	Retrieved text resolves to the uploaded document and its source type.	Historical JDs and question/answer guides remain distinguishable.	TC02.01

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR02.02	The system shall generate 768-dimensional embeddings through containerized Ollama and nomic-embed-text.	The model initializer verifies identity, dimensions and inference before backend readiness.	A missing model downloads once; cached models are reused.	TC02.02
FR02.03	The system shall retrieve relevant sources with ownership and source-type restrictions.	A JD search returns permitted JD references; a question search returns permitted question-bank material.	Similarity retrieves evidence; it is not a recruitment score.	TC02.03
FR02.04	The system shall isolate private CV vectors from shared knowledge and detect incompatible model changes.	Search never returns another application's private CV passages; incompatible vectors are not mixed.	Model changes require controlled re-indexing.	TC02.04

3.1.3 Job description preparation and approval

Actors: Interviewer. Interviewers may create a grounded draft conversationally or enter a job description manually, then review and approve a version.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR03.01	The system shall draft a JD using relevant historical sources and explicit interviewer details.	The draft contains grounded skills and responsibilities and records source references.	Unrelated roles cannot substitute for missing evidence.	TC03.01
FR03.02	The system shall ask for essential missing details instead of inventing requirements.	Insufficient source and user detail produces a concise follow-up without a fabricated draft.	No separate requirement-discovery approval workflow.	TC03.02
FR03.03	The system shall derive an internal screening profile and preserve required versus preferred criteria.	Saved JD criteria and the profile agree, including supported minimum experience.	Unknown qualifications, pay or benefits are not invented.	TC03.03

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR03.04	The system shall support manual creation, targeted edits and versioned approval.	Edits preserve unrelated content; explicit approval marks the reviewed version.	An approved version remains preserved when a new draft is made.	TC03.04
FR03.05	The system shall prevent generation or editing from implicitly publishing a job.	Drafting and approval alone produce no external post.	Publication is a separate confirmed operation.	TC03.05

3.1.4 Job publication through Discord

Actors: Interviewer, job posting MCP service. A reviewed approved job can be published to a configured Discord channel through a dedicated service.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR04.01	The system shall prepare a post from an approved JD version, application URL and deadline.	The preview displays the intended content, destination, application link and deadline.	Employer branding and channel are configured.	TC04.01
FR04.02	The system shall require explicit interviewer approval of the current preview before publication.	Unconfirmed or rejected previews do not produce Discord messages.	An outdated JD/preview requires renewed review.	TC04.02
FR04.03	The system shall publish through the private job posting MCP service and save its outcome.	A successful publication stores its external reference and reviewed version.	The bot token remains service-side.	TC04.03
FR04.04	The system shall handle repeated and uncertain publication requests safely.	Duplicate confirmation does not knowingly create another identical post; failures remain visible.	Do not blindly retry uncertain external outcomes.	TC04.04

3.1.5 Application intake and candidate management

Actors: Interviewer, Google Sheets and Drive integration. Google Form responses are imported read-only and routed to their intended job posting.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR05.01	The system shall import response rows and privately download supported CV/photo files.	Each imported application has the correct posting, source identity and document metadata.	Service-account viewer access and column mappings are required.	TC05.01
FR05.02	The system shall synchronize idempotently using source identities and file hashes.	Re-sync of an unchanged response does not duplicate applications or documents.	Failures identify affected rows without undoing successful imports.	TC05.02
FR05.03	The system shall isolate application profiles despite repeated email addresses.	Two applications with the same email cannot silently overwrite each other's identity or evidence.	Duplicate detection is conservative and traceable.	TC05.03
FR05.04	The system shall provide candidate search, job filters, pagination and private document access.	Counts and pages reflect the selected authorized posting/filter; CV access is authenticated.	Private storage paths and Drive URLs are not exposed publicly.	TC05.04
FR05.05	The system shall delete selected applications and prevent unintended re-import.	Deleted source responses are excluded on later sync unless explicitly repaired.	Dependent assessments and private artifacts follow deletion rules.	TC05.05
FR05.06	The system shall show job-specific notifications for pending form responses.	Pending counts direct the interviewer to the relevant posting and synchronization action.	Unavailable Google access is shown as unavailable, not zero pending.	TC05.06

3.1.6 CV extraction OCR and evidence screening

Actors: Interviewer, screening services. Screening extracts usable text and checks job requirements against source-backed candidate evidence.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR06.01	The system shall extract text from supported PDF and DOCX CVs.	Native documents retain page/block locators and original source text.	Default CV bounds are 10 MB, 40 pages and 150000 characters.	TC06.01
FR06.02	The system shall use local OCR for scanned PDF pages when native text is insufficient.	Accepted OCR retains page references and uncertainty warnings.	Default OCR cap is 8 pages, confidence 80 and uncertain ratio at most 0.15.	TC06.02
FR06.03	The system shall reject unreadable or unreliable extraction without awarding a zero score.	The affected document shows a processing issue and a retry path.	A processing failure is not evidence of candidate inability.	TC06.03
FR06.04	The system shall match approved requirements using candidate-specific retrieved passages.	Findings cover each requirement and contain verifiable source references.	Semantic equivalence does not imply equivalence of different named technologies or degree levels.	TC06.04
FR06.05	The system shall independently validate evidence and perform bounded recovery of unsupported findings.	Unknown quotations/passages are rejected; unresolved errors remain processing failures.	No protected-trait or institution-prestige preference.	TC06.05
FR06.06	The system shall reuse compatible results and report partial-batch outcomes.	Unchanged validated inputs reuse results; successful CVs remain usable when another fails.	Changed documents, criteria or scoring inputs invalidate stale results.	TC06.06

3.1.7 CV scoring and ranking

Actors: Interviewer, scoring services. Application code calculates explainable CV scores after evidence validation, rather than asking an LLM to invent totals.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR07.01	The system shall calculate requirement coverage using met 1, partially met 0.5 and unmet 0.	A fixed set of validated findings and approved weights yields the same exact total.	Unprocessed or failed documents remain unranked.	TC07.01
FR07.02	The system shall validate interviewer-approved CV category and requirement weights.	Category weights total 100 and invalid, negative or overprecise values are rejected.	Defaults are skills 50%, experience 30% and qualifications 20%.	TC07.02
FR07.03	The system shall handle categories without weighted requirements explicitly.	An empty category has no unsupported contribution; redistribution follows the approved policy.	Effective weights and the configuration are saved.	TC07.03
FR07.04	The system shall retain scoring snapshots, breakdowns, explanations and evidence.	Each ranked score traces to its requirement configuration and validated CV run.	Embedding similarity is not used as a candidate percentage.	TC07.04
FR07.05	The system shall use exact arithmetic and stable shared ranks for ties.	Equal exact totals produce ranks such as 1, 1, 3; display rounds half-up to two decimals.	Pagination and filtering do not renumber saved ranks.	TC07.05

3.1.8 Interview scheduling invitations and rescheduling

Actors: Interviewer, scheduling MCP service, SMTP. Manual controls and conversational commands use the same reviewed scheduling workflow.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR08.01	The system shall select candidates from saved rankings by checked application, name, top N or threshold.	A named candidate resolves within the selected posting; ambiguity requests clarification.	Top N is not silently replaced by lower-ranked excluded candidates.	TC08.01
FR08.02	The system shall distinguish not scheduled, scheduled and completed interviews.	The candidate selection view shows the relevant state and schedule details.	Duplicate active invitations are blocked.	TC08.02
FR08.03	The system shall prepare sequential individual slots and a recipient/time/email preview.	The preview shows each recipient, local time, duration, message and calendar attachment.	Asia/Dhaka is the default display/scheduling zone; times persist in UTC.	TC08.03
FR08.04	The system shall send SMTP invitations with private join links and ICS files only after confirmation.	Only the explicitly confirmed preview is delivered and its outcome is stored.	No Google/Outlook calendar-booking integration is required.	TC08.04
FR08.05	The system shall represent pending, sending, sent, failed and uncertain delivery outcomes.	Progress and per-recipient states remain accurate after refresh.	Uncertain delivery is not automatically resent.	TC08.05
FR08.06	The system shall support reviewed rescheduling and preserve schedule history.	Confirmed rescheduling updates slots and links while retaining prior history.	The scheduling MCP service forwards delegated, owned backend operations.	TC08.06

3.1.9 Live interview rooms and participant consoles

Actors: Interviewer, candidate, LiveKit Cloud. A private room supports device checks, waiting, admission and controlled one-to-one media.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR09.01	The system shall provide device checks and an interviewer-controlled waiting/admission flow.	A candidate cannot obtain joined-room capabilities before admission.	A candidate account is not required.	TC09.01
FR09.02	The system shall issue scoped room tokens and use direct browser WebRTC through LiveKit.	Only authorized participants can join the intended room.	Website/API tunneling is not the media transport.	TC09.02
FR09.03	The system shall support microphone, camera, leave, reconnect and end-interview controls.	Controls preserve room state and closure prevents new participant access.	Ending requires interviewer authority.	TC09.03
FR09.04	The system shall show a main participant view with a switchable smaller self-preview.	Selecting the preview swaps the displayed focus without disconnecting media.	Controls remain compact, labeled and keyboard usable.	TC09.04
FR09.05	The system shall open the interviewer console on candidate information and restrict candidate tools.	Interviewer sees profile/CV context first; candidate sees participation and practical work only.	No interview audio/video recording is stored.	TC09.05

3.1.10 Consented live transcription

Actors: Interviewer, candidate, transcription worker, Deepgram. An optional worker persists only final speech text with trustworthy speaker and timing metadata.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR10.01	The system shall require both participant consents and interviewer start before transcription.	Missing consent prevents a new transcription session; withdrawal stops authorized processing.	Consent must not be inferred from merely joining.	TC10.01

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR10.02	The system shall subscribe only to authorized participant microphone tracks.	Other participant, screen-share and untrusted tracks are ignored.	Two microphone streams are handled independently.	TC10.02
FR10.03	The system shall persist final segments idempotently with speaker, sequence and timestamps.	Duplicate/replayed final events do not create duplicate source evidence.	Interim text is not the durable transcript.	TC10.03
FR10.04	The system shall protect worker callbacks and keep transcripts interviewer-only.	A separate worker secret is required and public proxy access to internal callbacks is blocked.	Deepgram credentials never reach browsers.	TC10.04
FR10.05	The system shall expose gaps, provider failure and reconnect states without ending the human call.	Missing-speech warnings remain visible and saved final text is recoverable.	Default final-text retention is 30 days; no raw audio/video is retained.	TC10.05

3.1.11 Central and private interview assistants

Actors: Interviewer, scoped supervisor tools. One supervisor definition coordinates authorized business operations in different capability scopes.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR11.01	The system shall select and execute at most one business tool for each assistant request.	Malformed, unknown or multiple tool calls do not execute unauthorized operations.	The final response reports the actual tool result.	TC11.01
FR11.02	The system shall expose only the allowed central recruitment or private interview tools.	Room chat cannot publish, email, schedule, edit JDs or control meetings.	Server scope binds one authorized interview.	TC11.02

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR11.03	The system shall bind commands and approvals to the latest authentic interviewer message.	Retrieved content and fabricated tool arguments cannot supply approval.	Job/candidate IDs resolve only against authorized resources.	TC11.03
FR11.04	The system shall support context-aware follow-ups and named-candidate selection.	Confirmed details persist across related messages; names do not require a top-N cutoff.	Ambiguous resources require clarification.	TC11.04
FR11.05	The system shall ground interview-question suggestions in the question bank and JD.	Each suggested question names a relevant requirement and cites allowed source IDs.	No invented sources, discriminatory questions or unsupported candidate claims.	TC11.05

3.1.12 Transcript analysis and candidate performance

Actors: Interviewer, analysis services. Saved final transcript evidence and approved reference guides support granular answer assessment.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR12.01	The system shall detect substantive questions and pair them with candidate answers.	Questions, continuations, answers and clarifications remain distinguishable and reviewable.	Transcription session gaps and ambiguous boundaries are visible.	TC12.01
FR12.02	The system shall exclude housekeeping from scoring while retaining job-related nontechnical questions.	Greetings/device checks do not enter the score denominator; HR/sales/behavioral competencies can.	An off-topic answer to a valid job question is not excluded as small talk.	TC12.02
FR12.03	The system shall interpret context-supported ASR errors without fabricating speech.	Original quotations remain exact; meaningful interpretations have supporting context.	Competing grade-changing readings remain uncertain.	TC12.03

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR12.04	The system shall assign supported 0 to 100 marks for relevance, accuracy, completeness and quality.	Each dimension contributes 25%; numeric marks are not fixed by met/partial/unmet labels.	Correctness requires approved guides or technical references.	TC12.04
FR12.05	The system shall award zero for verified off-topic and genuinely unanswered questions only.	A verified mismatch yields zero; missing transcript, references or provider response does not.	Uncertainty and failures remain incomplete.	TC12.05
FR12.06	The system shall independently check quotations, interpretation, mappings and score support.	Unsupported evidence is rejected; requirement coverage preserves unassessed states.	No personality, accent, emotion or protected-trait inference.	TC12.06
FR12.07	The system shall save scores during Analyze answers and show concise traceable summaries.	Saved scores, strengths, areas to improve, gaps and verification needs are available without regrading.	Compatible accepted answers are reused; changed inputs become stale.	TC12.07

3.1.13 Live Work Sample Studio

Actors: Interviewer, candidate, practical review services. A short observed work product supplements verbal and CV evidence for technical and nontechnical roles.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR13.01	The system shall suggest job-grounded task choices and create only the selected private draft.	The interviewer selects a task and is directed to practical review before sharing.	Every task must be feasible in 1 to 15 minutes.	TC13.01
FR13.02	The system shall require task review, explicit sharing, candidate readiness and interviewer start.	A draft is private; working time starts only through the authorized state transition.	No generated criteria editor, model answers or required written explanation field.	TC13.02

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR13.03	The system shall enforce server-side timing, autosave and one immutable submission.	A second submission is rejected; expiry freezes the latest saved response with a clear status.	Candidate clock changes cannot extend the deadline.	TC13.03
FR13.04	The system shall show the interviewer the latest server-saved work.	Saved progress and revision status are available during the task.	Verbal explanation is requested by the interviewer, not a text field.	TC13.04
FR13.05	The system shall obtain an LLM correctness score and independent evidence-support check.	A defensible 0 to 100 result cites exact submitted text and line ranges.	Code is not executed; no plagiarism or AI-use detection is claimed.	TC13.05
FR13.06	The system shall count only the interviewer-approved verified AI score.	Approval preserves the proposed numerical result; pending/failed review is not zero.	The interviewer does not enter custom marks or automatically award 100.	TC13.06
FR13.07	The system shall support focused engineering, HR, sales and other role tasks.	A technical correction and a nontechnical scenario pass the same lifecycle checks.	Missing materials or task ambiguity may prevent a defensible score.	TC13.07

3.1.14 Optional photo verification and candidate profile

Actors: Candidate, interviewer, local identity service. The expected application's uploaded photograph supports optional local one-to-one comparison.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR14.01	The system shall display the authorized application's uploaded photo in candidate information.	A useful centered portrait loads privately or a fallback avatar appears.	Missing photos must not block application intake or interviews.	TC14.01

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR14.02	The system shall require in-app candidate consent before face-template processing and comparison.	Unchecked consent does not block joining; withdrawal removes authorized derived face data.	Google Form photo upload is not biometric consent.	TC14.02
FR14.03	The system shall let only the interviewer request a short, versioned face capture.	Capture uses three temporary samples from the existing camera and expires after 45 seconds.	Consent, active session, camera availability and reference version are checked.	TC14.03
FR14.04	The system shall compare locally and expose outcomes only to the interviewer.	Matched, unverified, missing, declined and manual states are distinguishable.	No legal-ID, liveness, cheating or continuous-monitoring claim.	TC14.04
FR14.05	The system shall encrypt references/templates and expire them under the configured policy.	Ciphertext is stored privately; the earlier of 30 days or 7 days after interview governs defaults.	Matching never affects scores or admission; original Drive upload follows its owner's policy.	TC14.05

3.1.15 Final score consolidation and ranking

Actors: Interviewer, final ranking service. Final ranking combines compatible saved assessments without launching new grading.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR15.01	The system shall combine interview 60%, CV 30% and approved practical 10%.	Interview 80, CV 70 and practical 90 produce 78.00%.	All component values are percentages under compatible saved policies.	TC15.01

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR15.02	The system shall read current saved component scores when Calculate final ranking is selected.	The calculation makes no fresh interview/practical grading request.	Missing or stale components direct the interviewer to the required action.	TC15.02
FR15.03	The system shall withhold a final rank when required components are incomplete.	Pending/unverified practical review, failed analysis and stale evidence do not become zeros.	Final weights are not silently redistributed.	TC15.03
FR15.04	The system shall preserve exact totals, ties and saved ranking identity.	Equal totals share ranks; search and paging retain authoritative ordering.	Displayed two-decimal percentages do not decide ties.	TC15.04
FR15.05	The system shall preserve historical policies and identify outdated reports.	Old totals remain traceable and incompatible policy versions are not combined silently.	No autonomous hiring or rejection decision.	TC15.05

3.1.16 Candidate reports evidence and exports

Actors: Interviewer, report services. Private reports summarize saved evidence and assessment components for review.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR16.01	The system shall provide concise performance details with optional traceable source inspection.	Strengths, weaknesses, evidence gaps and verification needs link to supporting saved evidence.	Summary presentation must not create a new score or unsupported claim.	TC16.01
FR16.02	The system shall generate private HTML/PDF candidate reports and ranking exports.	Authorized downloads preserve component scores, policy versions and evidence references.	Candidates and other owners cannot download the artifact.	TC16.02

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR16.03	The system shall render reports with scripts and external resource fetching disabled.	A malicious stored passage cannot execute scripts or fetch arbitrary network resources.	Export processing is isolated from normal request handling.	TC16.03
FR16.04	The system shall invalidate or mark reports stale when underlying evidence changes or expires.	Source versions, expiry and export status are visible and regeneration is explicit.	Opening, filtering and downloading do not trigger answer regrading.	TC16.04

3.1.17 Dashboard notifications and workflow navigation

Actors: Interviewer. The workspace exposes actionable recruitment progress without making consequential decisions automatically.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
FR17.01	The system shall summarize authorized jobs, application activity and upcoming interviews.	Dashboard cards and counts agree with saved resources and active filters.	Partial provider unavailability has an explicit state.	TC17.01
FR17.02	The system shall reveal the section produced by an explicit user action.	Drafting, scheduling previews, selected practical tasks and requested results become visible.	Background polling must not repeatedly steal focus.	TC17.02
FR17.03	The system shall preserve context across navigation and private assistant panels.	Selected job/candidate state and intended deep links remain consistent.	Do not replace authorized resource selection with guessed IDs.	TC17.03
FR17.04	The system shall show actionable loading, empty, error and retry states.	Failed requests identify the affected workflow and preserve saved work.	External side effects require confirmation despite UI retries.	TC17.04

4 User Interface

This section inventories the principal screens and their functional relationships. Section 8 retains the template's second User Interface section and defines cross-screen interaction and navigation behavior.

UI No	UI name	Related FR	Description	Test case ID
UI01	Sign in	FR01.01 to FR01.02	Interviewer credentials, validation and safe access errors.	TCUI01
UI02	Dashboard and notifications	FR05.06; FR17.01	Recruitment totals, pending form responses and upcoming interviews.	TCUI02
UI03	Jobs and JD review	FR03.01 to FR04.04	Job cards, draft/review, approved versions and Discord preview.	TCUI03
UI04	Candidates	FR05.01 to FR05.06	Search, job filtering, application selection, CV access and deletion.	TCUI04
UI05	CV Screening	FR06.01 to FR07.05	Weights, processing progress, CV ranking and requirement evidence.	TCUI05
UI06	Schedule interviews	FR08.01 to FR08.05	Eligibility/status, selection, slot preview and invitation delivery.	TCUI06
UI07	Scheduled interviews	FR08.06; FR09.01	Upcoming/completed sessions, rescheduling and console access.	TCUI07
UI08	Interviewer console	FR09.03 to FR14.05	Video; candidate info first; private assistant, transcript, analysis and practical tabs.	TCUI08
UI09	Candidate interview console	FR01.03 to FR01.05; FR09.01 to FR10.01	Private-link entry, device checks, compact consent, waiting, call and task.	TCUI09
UI10	Practical exercise	FR13.01 to FR13.07	Private draft, sharing, countdown, saved response, AI review and approval.	TCUI10
UI11	Assessment and reports	FR15.01 to FR16.04	Saved components, missing-score actions, final ranking and private exports.	TCUI11
UI12	Central assistant panel	FR11.01 to FR11.04; FR17.02	Contextual workflow chat, authentic confirmations and action-result navigation.	TCUI12

Screen names describe user-facing workspaces; nested tabs and panels need not be separate routes. Candidate-only controls must remain distinct from interviewer controls even when they share media components.

5 Non Functional Requirements

The following requirements govern cross-cutting behavior. Numerical performance values are acceptance objectives for a controlled test, not measurements already achieved or external-provider guarantees.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
NFR01	The system shall keep lengthy AI, import, export and screening work out of synchronous page rendering.	A request obtains a durable workflow ID/status and the page can show later progress.	Provider latency is external; background work still consumes host resources.	TCN01
NFR02	The system shall preserve accepted work across transient failures and page refreshes.	Saved drafts, final transcript segments, submissions and results can be retrieved after reconnect.	In-memory jobs need defined recovery after process restart.	TCN02
NFR03	The system shall report unavailable, partial, stale and failed states accurately.	A failed integration does not appear as zero candidates, a successful score or delivered mail.	Do not expose raw provider responses or secrets.	TCN03
NFR04	The system shall make read-only views free of hidden paid grading or external writes.	Opening, filtering and refreshing existing results produce no new assessment or publication.	Explicit live-update mode may analyze changed inputs under its bounded policy.	TCN04

5.1 Performance Requirements

Reference measurement profile: a warmed deployment with 4 vCPU and 8 GB RAM, 10 UI clients, 100 saved applications for one posting, and one active one-to-one interview. Record host, database, browser, network and provider conditions. These targets must be validated before claiming production capacity; initial model download and cold-start work are excluded.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
PER01	Authorized paginated reads shall target p95 response time of at most 2 seconds.	Measure 100 reads with 10 UI clients and 100 applications on the reference host.	Excludes cold start, external AI/media and large downloads.	TCP01
PER02	Background-work submission shall target acknowledgement within 2 seconds.	Measure 20 valid submissions; each returns a run ID or an actionable refusal.	Queue acknowledgement is not completion.	TCP02
PER03	Final ranking of 100 compatible saved applications shall target completion within 5 seconds.	Time request to available result and verify that no grading provider is called.	Database/ network latency must be recorded.	TCP03
PER04	Explicit UI actions shall provide visible feedback within 300 milliseconds.	Interaction tests show loading, disabled action or an immediate result before completion.	Excludes browser/OS permission dialogs.	TCP04
PER05	External work shall have bounded timeouts and visible progress or failure.	Inject stalled OCR, model and transcription calls; no workflow stays falsely ready.	OCR default timeout is 120 seconds; provider limits are configurable.	TCP05

5.2 Safety Requirements

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
SAF01	The system shall keep hiring and rejection decisions with the interviewer.	No score, photo outcome or assistant action autonomously hires or rejects a candidate.	Scores are decision support, not a guarantee of competence.	TCS01
SAF02	The system shall distinguish missing evidence from demonstrated poor performance.	Provider failure and uncertain speech remain incomplete; verified off-topic/unanswered answers can score zero.	Applies to all displayed and exported results.	TCS02

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
SAF03	The system shall preserve evidence and communicate uncertainty in AI output.	A questionable claim or ASR repair cannot become a definitive fact without cited support.	Perfect transcription and perfect model accuracy are not promised.	TCS03
SAF04	The system shall avoid protected-trait, emotion, accent and personality inferences.	Review prompts and fixtures; such attributes do not enter recruitment scoring.	Only job-related supplied evidence is assessed.	TCS04
SAF05	The system shall make face verification optional and non-consequential.	Declined, failed or inconclusive matching does not block admission or change a score.	Not legal identity, liveness, cheating or continuous-monitoring verification.	TCS05

5.3 Security Requirements

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
SEC01	The system shall enforce authenticated ownership on private business records and files.	Cross-owner and candidate access tests fail closed for records, transcripts and exports.	Every endpoint and tool must enforce scope.	TCSEC01
SEC02	The system shall protect credentials, signing secrets and provider keys server-side.	No secret appears in browser bundles, exported reports, repository fixtures or safe error messages.	Operators provision and rotate credentials separately.	TCSEC02
SEC03	The deployed service shall use HTTPS, restricted origins and appropriately secure session cookies.	Inspect deployed headers/cookies; altered, expired and revoked sessions fail.	Local development HTTP is not a production security profile.	TCSEC03

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
SEC04	The system shall treat uploaded documents, transcripts and retrieved passages as untrusted data.	Embedded instructions cannot authorize posting, invitations or privileged tool calls.	Consequential operations require authentic human confirmation.	TCSEC04
SEC05	The system shall bound file types, size, pages, images and generated output.	Oversized or malformed inputs produce safe errors before expensive processing.	Configured limits and supported MIME types apply.	TCSEC05
SEC06	The system shall isolate internal callbacks, worker secrets and MCP services from public routes.	Public proxy requests to internal callback paths fail; invalid worker secrets are rejected.	Private service networking does not replace application authorization.	TCSEC06
SEC07	The system shall encrypt biometric processing material and avoid retaining capture frames.	Database inspection shows encrypted reference/template fields; temporary samples are discarded.	Encryption-key loss requires controlled re-enrollment; backups remain sensitive.	TCSEC07
SEC08	The system shall sanitize report content and minimize sensitive logging.	Malicious report HTML cannot run script or fetch external URLs; logs omit candidate text and tokens.	Diagnostic access is restricted to operators.	TCSEC08

6 Design Constraints

These constraints preserve the current architecture and trusted boundaries. They do not require a rewrite into independent autonomous agents or replacement of existing integrations.

DC ID	Design constraint	Rationale / verification
DC01	Keep a modular FastAPI backend with shared domain services; frontend actions, MCP delegates and assistant tools use the same authorization and state rules.	Inspect routes, service boundaries and equivalent-flow tests.
DC02	Use PostgreSQL/pgvector for supported deployment persistence, with Alembic migrations and private persistent storage.	Preserve source relationships, 768-dimension vector compatibility and upgrades.
DC03	Use containerized Ollama with nomic-embed-text, a one-shot initializer, pinned identity/dimension checks and persistent cache.	Do not mix embeddings from incompatible models or require host Ollama in Docker mode.
DC04	Use the seven graph definitions listed below; direct JD RAG remains a service.	Retired requirement-discovery code is not a separate active workflow.
DC05	Route confirmed publication through job-posting MCP and confirmed scheduling through interview MCP/backend services.	MCP services do not own independent business databases.
DC06	Keep WebRTC media browser-to-LiveKit and optional transcription in a separate authorized worker.	No backend video proxy, no stored A/V recording and no media-over-website-tunnel assumption.
DC07	Assess code as text with source evidence; never execute applicant code as part of this baseline.	Avoid unsupported sandbox/security and functional-execution claims.
DC08	Treat local identity inference as optional, versioned and application-specific.	No global face search, legal-identity assertion or recruitment-score input.
DC09	Keep all score arithmetic in application code after evidence validation.	Generation proposes supported findings/marks; it cannot silently choose totals or final weights.
DC10	Do not implement candidate signup, an external calendar API or a new asynchronous broker solely to satisfy this SRS.	Those systems are outside the agreed current baseline.

Active graph definition	Purpose
HireSenseSupervisorGraph	Scoped central/private assistant routing, one allowed business tool and truthful result.
JobPostingAssistantGraph	Reviewed job-publication conversation and confirmed posting flow.
ScreeningWorkflow	CV extraction/OCR, requirement matching and source validation.
CandidateScoringWorkflow	Deterministic CV scoring and ranked assessment presentation.
InterviewAnalysisGraph	Question/answer detection, scoped assessment and evidence checks.
ConsolidatedAssessmentGraph	Combine compatible saved CV, interview and practical components.
WorkSampleGraph	Job-grounded practical-task suggestions and selected private draft.

The private interview assistant is a restricted capability scope of the supervisor definition. Practical correctness/review services do not constitute an additional LangGraph definition. Separate graph names alone do not imply a fully autonomous multi-agent system.

Tool scope	Allowed business tools
Central recruitment (9)	generate_job_description; edit_job_description; approve_job_description; job_posting; sync_responses; screen_candidates; interviews; candidate_assessments; help.
Private interview (4)	suggest_interview_questions; interview_analysis; suggest_practical_tasks; help.

7 Software Quality Attributes

Quality attributes apply to the complete workflow and are evaluated alongside feature-level acceptance. Test evidence must identify its environment, data and integration substitutions.

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
QUA01	The system shall present concise, understandable recruitment results.	A reviewer can identify score, status and next action without expanding detailed evidence.	Evidence remains available on demand.	TCQ01
QUA02	The system shall support keyboard-operated controls and non-color status cues.	Exercise tab order, focus visibility, button names, field labels and validation with a keyboard.	Use accessible patterns; no unverified compliance certification is claimed.	TCQ02
QUA03	The system shall preserve exact deterministic totals once evidence and policy are fixed.	Repeated arithmetic on the same validated inputs yields identical totals, ties and ranks.	New LLM assessments may vary; reuse compatible accepted runs.	TCQ03
QUA04	The system shall maintain shared business logic across UI and assistant entry points.	Equivalent authorized requests follow the same selection, preview, approval and state rules.	Avoid separate scoring or scheduling logic inside chat routes.	TCQ04
QUA05	The system shall support repeatable automated verification with isolated external services.	Default tests use doubles/fixtures; provider integration tests require explicit opt-in.	A test mapping in this SRS is not evidence of a passed test run.	TCQ05
QUA06	The system shall package its local dependencies and preserve persistent state across recreation.	A configured Docker deployment reuses PostgreSQL, storage and the downloaded embedding model.	External credentials, connectivity and suitable resources remain required.	TCQ06

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
QUA07	The system shall preserve versioned auditability of generated and reviewed artifacts.	A result resolves to its source hashes, policy/model version and relevant approval state.	Historical output may be retained but cannot masquerade as current.	TCQ07

8 User Interface

Interaction and navigation requirements complement the screen inventory in Section 4. Result-reveal behavior should help users see what an explicit action produced while preserving focus during passive updates.

UI No	UI name	Related FR	Description	Notes	Test case ID
UX01	Action-result reveal	FR17.02	After an explicit action, navigate to the owning page/tab and scroll the produced section into view.	Respect reduced motion and sticky-header clearance.	TCUX01
UX02	Stable background updates	FR17.02 to FR17.03	Polling refreshes status without repeated scrolling, focus theft or loss of selection.	Do not move a user reading another section.	TCUX02
UX03	Consistent scheduling state	FR08.01 to FR08.06	Show status text, time and an interview link; clarify ambiguity before selecting by name.	Filters and selected posting remain visible.	TCUX03
UX04	Compact permissions	FR10.01; FR14.02	Show one initially unchecked permission control with concise text and expandable processing details.	Unchecked joining remains possible; withdrawal APIs remain supported.	TCUX04
UX05	Candidate-first interview context	FR09.04 to FR09.05; FR14.01	Open candidate information first; keep the main call anchored left on desktop and secondary content scrollable.	Small screens reflow without hiding essential actions.	TCUX05
UX06	Role-restricted navigation	FR01.04; FR11.02	Candidate UI never offers transcripts, analysis, interviewer assistant or photo-match outcomes.	API authorization remains mandatory.	TCUX06

UI No	UI name	Related FR	Description	Notes	Test case ID
UX07	Readable assessment cards	FR12.07; FR16.01	Use short strengths/improvement highlights; expand gaps, verification needs and sources on demand.	Color/icons supplement text; no wall of explanations.	TCUX07
UX08	Visible scoring readiness	FR12.05 to FR12.07; FR15.03	Distinguish queued, analyzing, complete, incomplete, error, excluded and stale scores.	Never show incomplete as ready or failed as zero.	TCUX08
UX09	Safe confirmations and progress	FR04.02; FR08.05; FR13.02	Preview external actions; show recipient/task states and prevent repeated confirmation while processing.	An uncertain send is not presented as a confirmed failure.	TCUX09
UX10	Recovery without data loss	FR17.04; FR13.03	An error leaves saved work intact and offers a bounded retry or correct next action.	A practical submission cannot be reopened as a new attempt.	TCUX10

9 Other Requirements

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
OTH01	The system shall persist recruitment relationships with database integrity and migrations.	A fresh database migrates successfully; upgrade tests retain existing owned relationships.	PostgreSQL/ pgvector is the supported deployment database.	TCO01
OTH02	Operators shall back up the database, private storage and identity encryption key as one protected recovery set.	A documented restore recovers linked records/files; missing keys are identified before enrollment reuse.	Backup schedule and off-site location are deployment decisions.	TCO02
OTH03	The system shall expire retained transcript and practical content according to configured policy.	Maintenance removes expired sensitive content and marks dependent results/reports appropriately.	Default retention is 30 days for each; no interview A/V recordings are stored.	TCO03
OTH04	The system shall apply biometric withdrawal, application deletion and derived-data expiry consistently.	Withdrawal invalidates the match and deletes app-controlled biometric material; re-sync does not reenroll silently.	Original Google Drive uploads and external backups have separate owner policies.	TCO04
OTH05	The system shall preserve data provenance and UTC times with readable local presentation.	Evidence resolves to an application and source/version; Dhaka display round-trips to the stored instant.	Source deletion/expiry is an explicit state, not fabricated evidence.	TCO05
OTH06	The system shall provide clear provider-quota failures and avoid unrequested paid reanalysis.	Quota/authentication/ timeouts yield safe retry guidance; read-only views do not spend inference quota.	Token/cost estimates depend on workload and configured provider pricing.	TCO06

Req. ID	Description	Acceptance / completion criteria	Limitations / constraints	Test case ID
OTH07	The project shall use authorized or clearly identified synthetic data for demonstrations and tests.	Demo fixtures contain no real applicant secrets; a full flow is traceable to its data source.	Synthetic outputs must not be represented as measured real-world accuracy.	TCO07
OTH08	The deployed system shall document prerequisites, operational limits and acceptance evidence.	Setup and relevant feature guides identify credentials, models, services, limits and verification commands.	This SRS does not certify production readiness or legal compliance.	TCO08

9.1 Data and retention baseline

Data group	Stored information	Retention / deletion boundary
Jobs, applications and assessments	Owned records, source response IDs, approved versions, findings, score/configuration snapshots.	Until authorized deletion or applicable operational retention; do not invent a global indefinite policy.
Private CVs and knowledge	Authorized files/chunks, page/block locators and vectors.	Deleted or invalidated with their owning record and applicable policy; vectors are not shared across candidate scope.
Transcript and practical content	Final text, source segments, submitted work and evidence.	Default 30 days each; expiry may invalidate dependent details/reports. No raw interview A/V recordings.
Biometric processing	Encrypted reference/templates; audit state and current consent.	Earlier of 30 days or 7 days after interview; withdrawal/app deletion removes app-controlled material. Temporary frames are not retained.
Original form uploads / email	Source Drive uploads and delivered messages are controlled outside HireSense.	Read-only integration cannot delete originals or revoke already delivered email; private links can be revoked.
Secrets, model cache and backups	Private keys, downloaded model assets and protected recovery copies.	Operator-controlled protection/rotation; model and encryption metadata must stay compatible.

9.2 Verification and traceability

For every TC identifier, the acceptance record should include requirement ID, setup/preconditions, authorized actor, input, action, expected result, observed result and pass/fail evidence. Use deterministic fixtures for arithmetic and adverse-state cases; use authorized or labeled synthetic records for end-to-end demonstrations.

The source files below are relevant implementation-test areas, not a one-to-one claim that every SRS acceptance scenario has already been automated or passed. Paths are under backend/tests/ unless prefixed with frontend/. Live-provider, PostgreSQL and browser-media checks require their documented environment.

No.	Acceptance scenarios	Existing verification area
01	TC01.01–TC01.05	test_auth.py; test_interview_rooms.py
02	TC02.01–TC02.04	test_rag.py; test_knowledge_pdf.py; test_ollama_bootstrap.py; test_semantic_screening.py
03	TC03.01–TC03.05	test_direct_job_description.py; test_job_description.py; test_manual_jobs.py; test_requirement_discovery_retirement.py
04	TC04.01–TC04.04	test_job_posting.py; test_mcp_config.py
05	TC05.01–TC05.06	test_google_forms_intake.py; test_application_identity.py; test_candidate_bulk_delete.py; test_google_forms_notifications.py; test_candidate_cv_access.py
06	TC06.01–TC06.06	test_cv_ocr.py; test_screening.py; test_semantic_screening.py
07	TC07.01–TC07.05	test_scoring_calculation.py; test_scoring.py; test_scoring_presentation.py
08	TC08.01–TC08.06	test_interviews.py; test_interview_tools.py; test_livekit_cloud.py
09	TC09.01–TC09.05	test_interview_rooms.py; test_livekit_cloud.py; test_livekit_integration.py
10	TC10.01–TC10.05	test_transcription.py; transcription_worker/test_worker.py
11	TC11.01–TC11.05	test_supervisor_graph.py; test_supervisor.py; test_supervisor_candidates.py; test_interview_copilot.py; test_interview_tools.py
12	TC12.01–TC12.07	test_interview_analysis.py; test_interview_scope.py; test_transcript_scoring_robustness.py; test_performance_summary.py
13	TC13.01–TC13.07	test_work_samples.py; test_work_sample_postgres.py

No.	Acceptance scenarios	Existing verification area
14	TC14.01–TC14.05	test_candidate_identity.py; test_candidate_photo.py; test_identity_postgres.py
15	TC15.01–TC15.05	test_final_ranking.py; test_assessments.py
16	TC16.01–TC16.04	test_assessments.py; test_performance_summary.py; test_final_ranking.py
17	TC17.01–TC17.04	test_google_forms_notifications.py; test_supervisor_candidates.py

Cross-screen verification also uses frontend/tests/rooms/, including account-access, candidates-page, schedule-interviews-page, invitation-delivery-status, console, candidate, analysis, identity, work-samples, final-ranking-table, reports-page and workflow-navigation tests.

9.3 End-to-end acceptance scenarios

Scenario	Required path	Pass condition
E2E01 — Technical role	Grounded/manual JD → approval → confirmed post → import native/scanned CVs → ranking → confirmed invitation → admitted interview → consented transcript → answer analysis → one task → approved AI review → final ranking/report.	Each transition is authorized; evidence remains traceable; final 60/30/10 arithmetic uses saved compatible scores.
E2E02 — Nontechnical role	HR/sales JD → applications → role-related questions → short business scenario task → reviewed assessment.	Relevant nontechnical competencies are included; greetings/logistics are excluded; task fits 15 minutes.
E2E03 — Degraded services	Inject OCR uncertainty, quota failure, missing references, uncertain mail and failed photo comparison.	Failures remain actionable, saved work survives and no missing result becomes an invented zero, success or hiring decision.
E2E04 — Privacy boundaries	Cross-owner access, candidate transcript/report requests, consent withdrawal, stale capture and expired private links.	Unauthorized access fails; withdrawal/revocation takes effect; candidate can still join without optional processing.
E2E05 — Reuse and change	Reopen saved scores; calculate ranking; change a source or policy; re-evaluate readiness.	Read-only use does not regrade; incompatible results are marked stale and require an explicit appropriate action.

9.4 Configuration and operational limits

Document limits are defaults, not candidate-selection rules: CV 10 MB, 40 pages and 150,000 characters; native-text minimum 80 characters; OCR up to 8 pages, confidence threshold 80, uncertain-text ratio at most 0.15 and 120-second timeout. Reference PDF ingestion has separate limits (15 MB and 100 pages) and does not imply automatic scanned-knowledge OCR. A screening batch is limited to 50 applications by default.

Practical tasks last 1–15 minutes. Identity capture requests expire after 45 seconds and require three temporary samples. Join-link/session/token lifetimes, model names, timeouts and retention are deployment configuration. Limits must be validated consistently at the backend and reflected in user-facing guidance.

External-provider tokens, rate limits and costs depend on workload and account tier. Use docs/llm-token-usage-estimate.md as an approximate planning model, not a billing guarantee. Retries, independent evidence checks, source length and changed inputs can increase usage. Containerized embedding inference is local, but consumes host compute.

9.5 Acceptance and change control

The team and supervisor should review this baseline, complete cover details and record any approved changes in Revision History. A change to scoring weights, evidence policy, privacy scope, provider/model compatibility or external side effects requires impact review and updated tests/documentation. Acceptance depends on observed evidence, not the presence of an AI-generated result or a documentation claim.

Remaining deployment decisions include the hosting security profile, lawful processing basis and local obligations, backup schedule, operational monitoring and measured capacity. These decisions do not authorize hidden candidate monitoring or expansion beyond the stated system boundary.